NET732

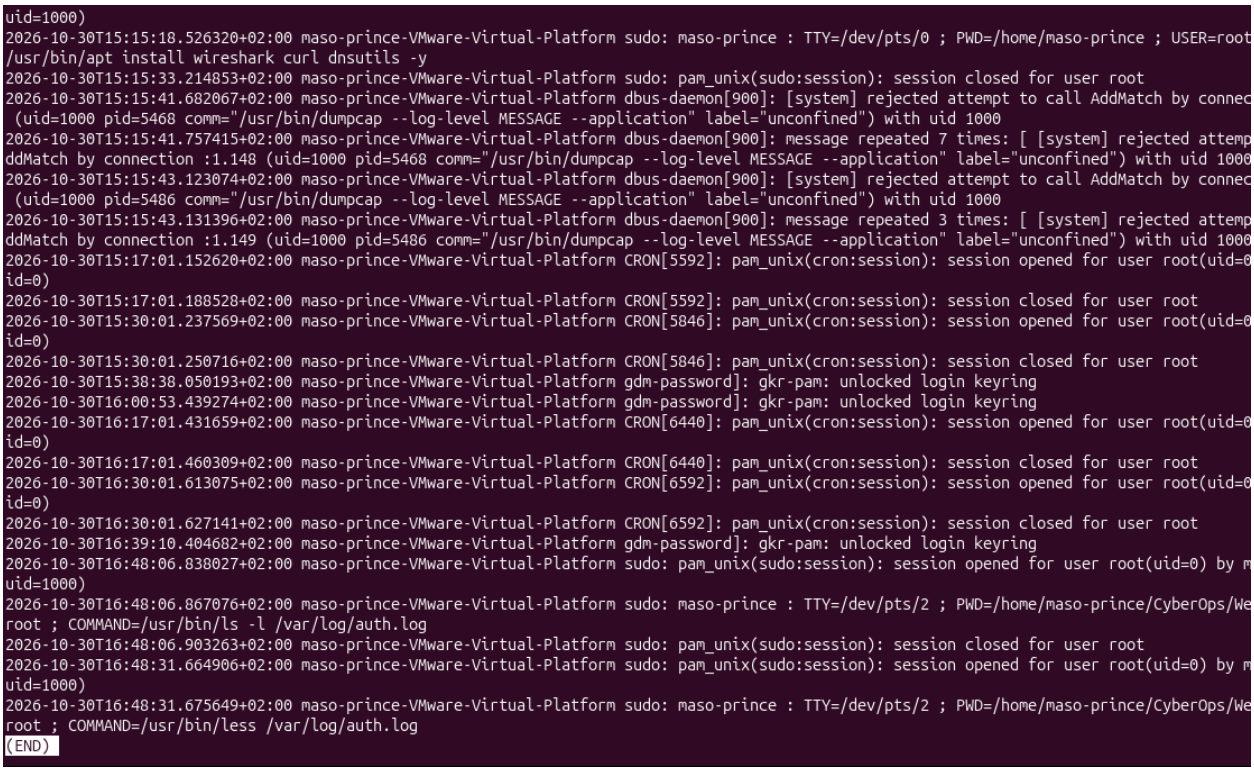
Week 4:Log Analysis

Practical

1. Accessing authentication logs

I opened Ubuntu Terminal and checked the authentication logs using:

sudo journalctl



I used this command to view system events and authentication activities.

2. Investigating failed login attempts

I searched for failed authentication attempts using:

sudo journalctl | grep -i "authentication failure"

I also checked for failed passwords:

sudo journalctl | grep -i "failed password"

```
maso-prince@maso-prince-VMware-Virtual-Platform:~/CyberOps/Week4$ sudo journalctl | grep -1 "authentication failure"
maso-prince@maso-prince-VMware-Virtual-Platform:~/CyberOps/Week4$ sudo journalctl | grep -1 "failed password"
maso-prince@maso-prince-VMware-Virtual-Platform:~/CyberOps/Week4$
```

This helped me look for unsuccessful login attempts that could indicate unauthorised access.

3. Checking login activity

I used the following commands:

last

who

```
maso-prince@maso-prince-VMware-Virtual-Platform:~/CyberOps/Week4$ last
maso-pri tty2         local            Fri Oct 30 14:16 - still logged in
maso-pri tty2         local            Fri Oct 30 13:45 - still logged in

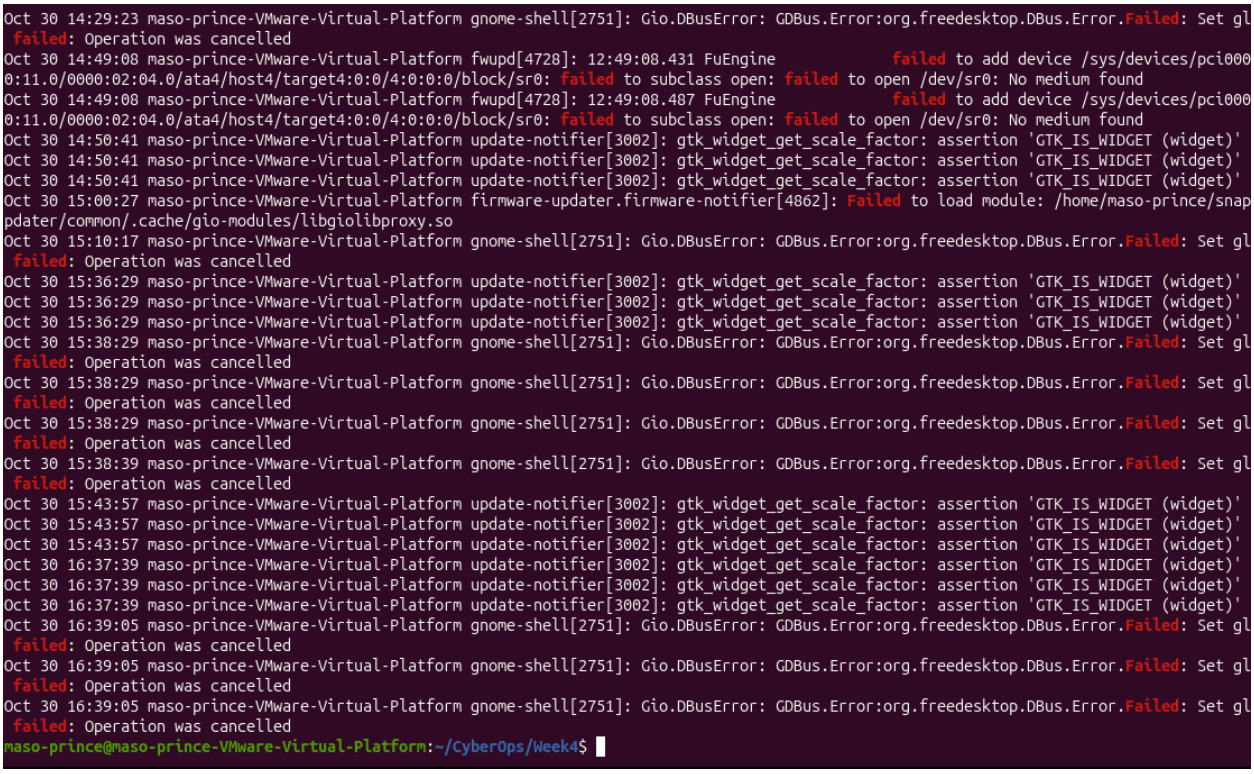
wtmpdb begins Fri Oct 30 13:45:13 2026
maso-prince@maso-prince-VMware-Virtual-Platform:~/CyberOps/Week4$ who
1000     seat0        2026-10-30 14:15
1000     tty2         2026-10-30 14:15
maso-prince@maso-prince-VMware-Virtual-Platform:~/CyberOps/Week4$
```

I used last to check previous login sessions and who to check currently logged-in users.

4. Identifying suspicious activity

I searched for possible suspicious events using:

sudo journalctl | grep -iE "failed|invalid user|authentication failure"



I used the results to check for failed logins, unknown usernames and repeated authentication failures.

5. Saving logs and evidence

I saved the system logs using:

sudo journalctl > authentication.log

I also used the script command to record my terminal activities in commands.log.

My Week 4 folder contains:

- authentication.log

- commands.log

6. Analysis and conclusion

During this practical, I learned how to investigate authentication logs using Ubuntu. I used different commands to check login activity and search for failed authentication attempts.

This is important in SOC operations because security analysts monitor logs to identify suspicious login activity and possible unauthorised access.

I successfully completed the log analysis activities and saved my evidence. My next step is to practise identifying repeated failed logins and investigating possible security threats.